Elmaz Abliazova

exa349@case.edu|(216) 303-2868
www.linkedin.com/in/elmaz-abliazova
https://elmazabliazova.github.io/#

EDUCATION

Case Western Reserve University **Expected May 2028**
Bachelor of Science in Biomedical Engineering |GPA: 3.55
Track: Biomechanics
Minor: Business Management
Relevant Coursework: Design and Manufacturing I&II, Statics, Dynamics, Strength of Materials, Biomedical Signals and Systems, Circuits and Instrumentation, Physiology-Biophysics I & II, Tissue Biomechanics
Awards: UWC Davis Scholarship, Dean's High Honors List

SKILLS

Technical Skills: SolidWorks; Fusion 360; 3D Printing (Prusa, Bambu Lab); Additive Manufacturing (FDM, PolyJet, Form 4); 3D Slicing Software (GrabCAD, PreForm, PrusaSlicer); Rapid Prototyping; Testing & Validation; Soldering & Electronics Assembly; MATLAB; Java; Laser Cutting; CNC Machining (Router, Mill, Lathe); Woodshop Tools; Tool Setup & Calibration; Part Finishing
Certifications: CSWA (Certified SolidWorks Associate)

PROFESSIONAL EXPERIENCE

Afference **June 2026 - Present**
Design Prototyping Intern Cleveland, OH

- Modeled and iterated on 8+ hardware component designs in SolidWorks and Fusion 360 to support future development.
- Fabricated wearable prototypes, integrating technical textiles, micro-wire soldering, and benchtop assembly techniques.
- Developed HITL protocols to identify optimal electrode placement for tactile stimulation, testing across 15 subjects.
- Designed and built 25 custom electromechanical test fixtures to standardize R&D hardware testing and validation.

Sears think[box] Makerspace **Jan 2026 - Present**
Prototyping Student Technician Cleveland, OH

- Operated additive manufacturing equipment across FDM, SLA, and PolyJet 3D printing technologies.
- Performed maintenance, calibration, and troubleshooting on 6 industrial 3D printers and other prototyping equipment.
- Led over 40 hands-on workshops training users in prototyping, engineering software, and digital fabrication tools.
- Developed 3 industry partnership projects by creating technical designs and fabrication files to meet requirements.

DESIGN PROJECTS

PlumBio Entrepreneurial Project **Jan 2026 - Present**

- Conducted research and designed SolidWorks prototypes for early-stage bioengineering devices.
- Developed and iterated on functional prototypes, testing form, function, and manufacturability.

Foldable Emergency Stretcher - Mechanical Manufacturing Project **Sep 2025 - Dec 2025**

- Designed a portable stretcher with a telescoping frame and locking pins for rapid deployment.
- Applied SolidWorks topology optimization to reduce crossbar weight by ~50% while maintaining rigidity.

Custom Gratnells Bin Storage System - Makerspace Partnership Project **July 2025 - August 2025**

- Generated 2D CAD layouts and toolpaths in VCarve to optimize panel layout and CNC machinability for a storage unit.
- Machined and assembled the unit on a Laguna CNC router, completing woodshop post-processing and assembly.

LEADERSHIP EXPERIENCE

Global Health Design Collaborative(Vaccine Carrier Team) **Sep 2024 - Present**
Technical Manager |Case Western Reserve University

- Led SolidWorks CAD redesigns that doubled vial capacity, improving ergonomics and thermal performance.
- Executed thermal loss trials across 3 prototype iterations, maintaining internal temperatures below 6°C for 12+ hours.
- Directed project alignment and technical decisions across the team while incorporating healthcare worker feedback.

Biodesign Club **Sep 2023 - Present**
Founder & President |Case Western Reserve University Cleveland, OH

- Direct technical strategy for a 100+ member club, guiding 3 teams developing novel biomaterials and lab devices.
- Developed baseline material formulation protocols, lab safety frameworks, and empirical testing procedures.
- Created project management systems, needs-identification workflows, and technical documentation.